
Speed

The Cost of Execution Delay

Three trading signals on a liquid stock, a less liquid stock and a currency pair

MFE 230X: High-Frequency Finance

Final Project, Part 1

Group 06

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Assets	AAPL, GPRO, EUR/USD
Period	September 2019, 20 trading days
Signals	Moving average, on-balance volume, persistent quote imbalance
Execution delays	0 ms, 100 ms, 1 second
Book	\$1,000,000

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Contents

1	Executive summary	2
2	Assignment and method	3
2.1	What was asked and what we chose	3
2.2	Trading procedure	3
2.3	Three measures of the cost of delay	3
3	Data and markets	4
3.1	Sources	4
3.2	Three very different markets	4
3.3	Limits of the data	4
4	The three signals	5
4.1	Moving average, MA (paper, equation 3)	5
4.2	On-balance volume, OBV (paper, equation 6)	5
4.3	Persistent quote imbalance, IMB (our own signal)	5
4.4	Parameters and rule universe	6
5	The 27 profit-and-loss reports	7
5.1	Where the P&L comes from	7
6	Cost of delay of the nine strategies	9
7	Importance of speed over a universe of 601 rules	11
7.1	All rules (Figure 6 of the paper)	11
7.2	Winning and losing rules, and the selection bias (Figure 7 of the paper)	12
7.3	Comparison with the paper	14
7.4	Effect of the delay per order	14
8	Answers to the questions of the assignment	16
8.1	Question (a): which asset is affected by delays, and why?	16
8.2	Question (b): do the signals depend on market conditions?	16
8.3	Question (c): does tick frequency matter, and how to use it?	17
9	Conclusion and limitations	19
9.1	Conclusion	19
9.2	Limitations	19
A	Rule universe	20
B	Code and reproduction	20

1 Executive summary

We apply the method of Scholtus and van Dijk (2012) to a liquid stock (AAPL), a less liquid stock (GPRO) and a currency pair (EUR/USD) over September 2019. A trading signal is computed every minute, and the order it implies is executed 0 ms, 100 ms or 1 second later at the best bid or ask of that moment. The question is how much return the delay costs.

Key finding. A delay of 100 ms costs nothing we can measure. A delay of 1 second costs 0.06 to 0.11 bps per order on the two stocks, and on AAPL it lowers the return of profitable rules by 3.1%, as in the paper. On EUR/USD delay has no cost. In every case the cost of delay is small next to the bid–ask spread.

Table 1. Results at a glance.

Question	Answer	Section
Do the strategies make money?	No. All 27 lose once the spread is paid, from -0.19% to -109% . On GPRO the signals earn money at the mid quote and the 22.5 bps spread takes it back.	5
Does a delay cost a single strategy?	Too small to measure over one month: $+0.02\%$ at 100 ms and -0.12% at 1 second on average, with a 90% interval of -0.42% to $+0.14\%$.	6
Does it cost over many rules?	Yes at 1 second on the two stocks: 0.06 to 0.11 bps per order. No at 100 ms: every effect is within 0.02 bps.	7.4
Does speed matter for profitable rules, as in the paper?	Yes on AAPL: -0.7% at 200 ms and -3.1% at 1 second, beyond the selection bias. Not shown on GPRO and EUR/USD.	7.2
Which asset is most exposed, and why?	AAPL, whose quote changes 8.8 times per second; then GPRO at 1 second, where each tick is worth 22 bps. EUR/USD is not affected.	8
Is a \$1,000,000 order realistic?	Only on EUR/USD. It is 22 times the displayed size on AAPL and 53 times on GPRO.	3

How the report is organised. Sections 2 to 4 describe what we implemented: the trading procedure, the data and the three signals. Section 5 gives the 27 profit-and-loss reports. Sections 6 and 7 measure the cost of delay, first for the nine strategies and then over a universe of 601 rules per asset. Section 8 answers the three questions of the assignment, and Section 9 concludes with the limitations.

2 Assignment and method

2.1 What was asked and what we chose

Table 2. The assignment and our choices.

Requirement	Our choice
Two equities, one liquid and one less liquid	AAPL and GPRO, on Nasdaq
One FX pair	EUR/USD
A one-month period	September 2019, 20 trading days
Two trading signals from the paper	Moving average (MA), on-balance volume (OBV)
One trading signal of our own	Persistent quote imbalance (IMB)
Execution delays	0 ms, 100 ms, 1 second
Total return on a \$1,000,000 book	27 reports: 18 asked for, 9 for our own signal (Section 5)
Cost of delay, Figure 6 of the paper	Sections 6 and 7
Comments on the effects observed	Section 8

2.2 Trading procedure

The procedure follows section 3.1 of the paper and the instructions of Discussion Session 06.

- **Signal.** Every 60 seconds, from 09:40 to 15:50 New York time, a rule computes a position in $\{-1, 0, +1\}$ from the information available at the interval change. The book is closed every day at the 15:50:00 price.
- **Execution.** The order is executed a delay d later at the best quote of that moment: buys pay the ask, sells receive the bid. The spread is the only transaction cost.
- **Size.** Every trade is for the number of units worth \$1,000,000 on the asset's first day: 4,850 AAPL shares, 263,505 GPRO shares, EUR 913,013.
- **Return.** The total return of a strategy is the sum of the simple returns of its round trips, not compounded. The dollar P&L is the same sum valued on the fixed number of units.

For a long position entered at t_1 and closed at t_2 , and for a short one, the round-trip returns are

$$r^{\text{long}} = \frac{p_{t_2+d}^{\text{bid}} - p_{t_1+d}^{\text{ask}}}{p_{t_1+d}^{\text{ask}}}, \quad r^{\text{short}} = -\frac{p_{t_2+d}^{\text{ask}} - p_{t_1+d}^{\text{bid}}}{p_{t_1+d}^{\text{bid}}}.$$

2.3 Three measures of the cost of delay

1. **Per strategy** (Section 6): the total return with delay d minus the total return with $d = 0$, for the same signal. **A negative number means that the delay hurts**, here and throughout the report.
2. **Over a universe of rules** (Sections 7.1 and 7.2): the paper's importance of speed, with its test of the selection bias for rules that make money and rules that lose money.
3. **Per order** (Section 7.4): the change in the execution price of each order caused by the delay.

3 Data and markets

3.1 Sources

- **Equities.** Databento Nasdaq TotalView-ITCH (MBP-1): every trade and every change of the best bid and offer, with nanosecond time stamps.
- **EUR/USD.** Dukascopy ticks: every change of the best bid and ask with the quoted sizes, with millisecond time stamps. The discussion session points to a currency file on the MFE-4 server; we use a public source so that the FX leg covers the same month, the same 20 days and the same clock window (13:30 to 20:00 UTC) as the equities.

3.2 Three very different markets

Table 3. The three markets, regular-session averages (units: shares or euros).

	AAPL	GPRO	EUR/USD
Mid price	217.90	4.51	1.1011
Quoted spread (bps)	0.73	22.52	0.27
One tick, in bps of the price	0.46	22.18	0.09
One-minute volatility (bps)	5.3	17.0	1.1
Quote price changes per second	8.80	0.04	1.21
Quote changed after 100 ms (%)	25.6	0.2	15.0
Quote changed after 1 s (%)	62.2	1.1	55.0
Units in a \$1,000,000 order	4,850	263,505	913,013
Units displayed at the best quote	225	4,945	1,673,822
Order size / displayed size	21.5	53.3	0.55

- **Cost of trading.** EUR/USD is the cheapest (0.27 bps quoted spread) and AAPL is close behind (0.73 bps). GPRO costs 22.5 bps: it trades near \$4.50, so the one-cent tick alone is 22 bps of the price and its spread is pinned at one tick.
- **Tick frequency.** AAPL's best quote changes price 8.8 times per second, EUR/USD 1.2 times, GPRO once every 25 seconds. After 1 second the quote has changed 62% of the time on AAPL, 55% on EUR/USD and 1.1% on GPRO.
- **Size available.** The \$1,000,000 order is 22 times the size displayed at the best quote on AAPL, 53 times on GPRO, and half of it on EUR/USD.

3.3 Limits of the data

- **No depth.** The paper and the discussion session use a price-impact price, the average price paid when the order walks the book. MBP-1 has the best quote only, so every order is assumed to fill in full at the best quote: realistic for the currency pair, optimistic for the two stocks. Returns in percent then do not depend on the order size.
- **No FX trade tape, one venue.** For EUR/USD the volume input of OBV is tick volume (one unit for each quote that moves the mid), and Dukascopy's tick frequency understates the whole market.

4 The three signals

Inputs are sampled at each one-minute interval change t . Let m_t be the mid quote and $\bar{x}_t(n)$ the average of the last n values of a series x . Figure 1 shows the three inputs and the positions they produce on one day.

4.1 Moving average, MA (paper, equation 3)

With a short window s , a long window l and a band b ,

$$\text{MA}_t = \begin{cases} +1 & \text{if } \bar{m}_t(s) > (1+b)\bar{m}_t(l), \\ -1 & \text{if } \bar{m}_t(s) < (1-b)\bar{m}_t(l), \\ 0 & \text{otherwise.} \end{cases}$$

Baseline: $s = 5$, $l = 30$, $b = 5$ bps. This is the smallest non-zero band of the paper's grid, chosen so that the rule also trades on EUR/USD.

4.2 On-balance volume, OBV (paper, equation 6)

OBV is the running sum since the open of trade volume v_j signed by the tick rule, auction crosses excluded:

$$\text{obv}_t = \sum_{j \leq t} \text{sign}(p_j - p_{j-1}) v_j.$$

The rule is long if $\overline{\text{obv}_t}(s) > \overline{\text{obv}_t}(l) + \text{band}$ and short if $\overline{\text{obv}_t}(s) < \overline{\text{obv}_t}(l) - \text{band}$. Baseline: $s = 5$, $l = 30$, $b = 0.5$. There are two departures from the paper.

- **Sign.** Volume is added on an uptick and subtracted on a downtick, the standard convention. The equation as printed in the paper has the opposite sign.
- **Band.** The paper's band is proportional to the level of OBV. Because OBV restarts at zero every day and changes sign, our band is b times the average one-minute volume over the long window.

4.3 Persistent quote imbalance, IMB (our own signal)

At every instant the imbalance of the best quotes is

$$\frac{q^{bid} - q^{ask}}{q^{bid} + q^{ask}},$$

with q the size at the best bid and ask. I_t is its time-weighted average over the minute ending at t . The rule is long if $I > \theta$ in each of the last k minutes, short if $I < -\theta$ in each of them, and flat otherwise. Baseline: $k = 2$, $\theta = 0.3$.

The idea. A queue that stays heavier on the bid for whole minutes signals buying pressure that has not yet moved the price. Requiring persistence filters the flickering that dominates instantaneous imbalance.

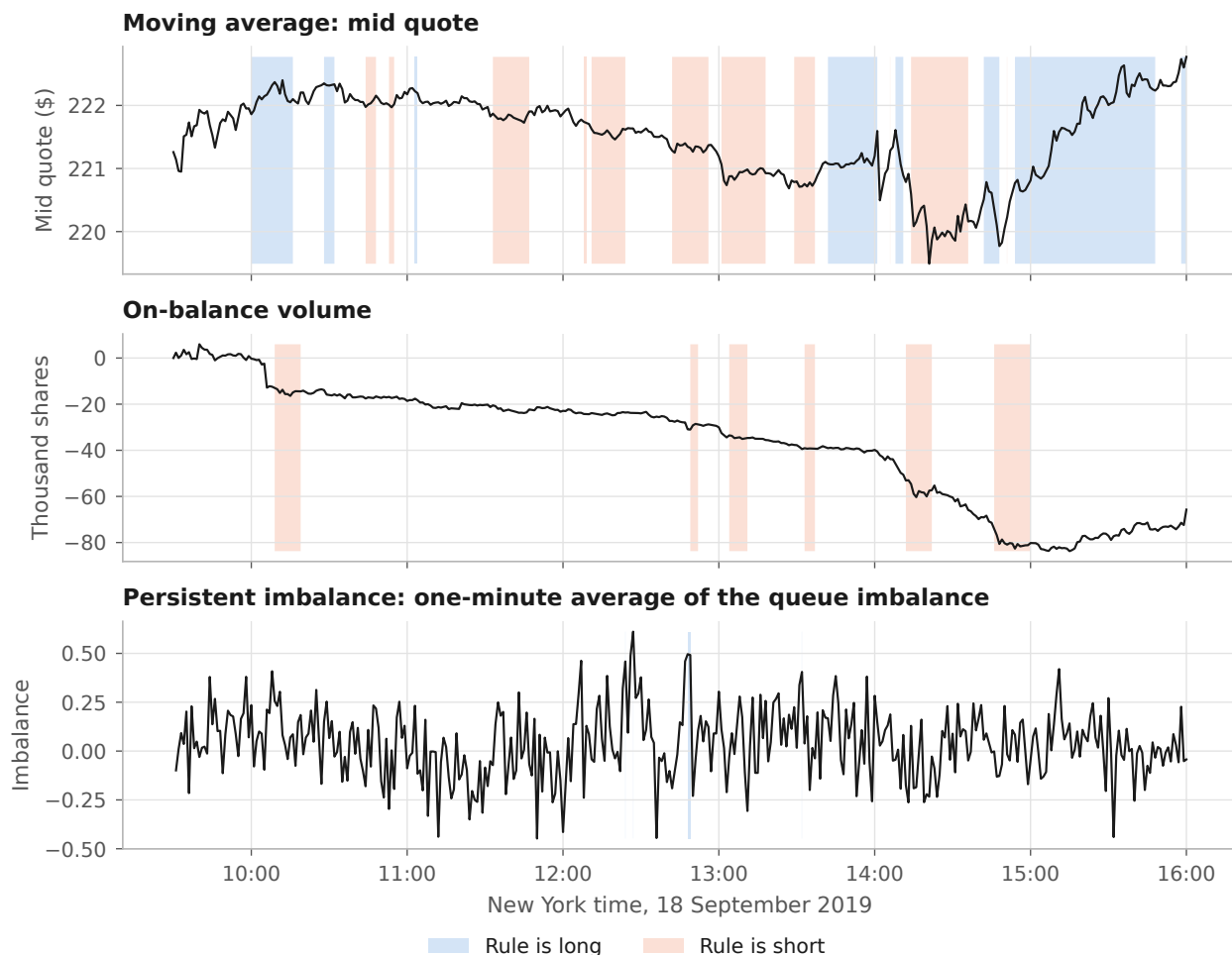


Figure 1. The three signal inputs on AAPL on 18 September 2019, and the positions taken by the baseline rules.

4.4 Parameters and rule universe

- **Parameters.** The baseline lookbacks and bands are values of the paper’s 60-second settings (its Appendix A). The same parameters are used for the three assets and were not tuned to improve the P&L.
- **Start of trading.** Lookback windows start at the 09:30 open, so the baseline MA and OBV give their first signal at 10:00 and IMB at 09:40. The paper warms its rules up on pre-market quotes; we do not, because GPRO’s pre-market book is too thin to give a meaningful mid.
- **Rule universe.** One rule per signal is enough for the 27 reports, but not to measure an effect as small as the cost of delay. For that part we follow the paper and use many rules: every combination of its 60-second grid for MA and OBV (288 rules each) and 25 IMB rules, 601 rules per asset in total (Appendix A).

5 The 27 profit-and-loss reports

Key finding. All 27 strategies lose money once the spread is paid. On GPRO the signals are right about the direction and the spread takes more than they earn; on AAPL the trend signal itself loses. The three delays give almost the same result.

Table 4. The 27 profit-and-loss reports. Total return is the sum of the simple round-trip returns; P&L is in dollars for the fixed number of units.

Asset	Signal	Trips	Total return (%)			P&L (\$)		
			0 ms	100 ms	1 s	0 ms	100 ms	1 s
AAPL	MA	304	−5.75	−5.88	−6.03	−60,963	−62,273	−63,922
	OBV	103	−0.89	−0.81	−0.90	−9,603	−8,827	−9,700
	IMB	42	−0.75	−0.77	−0.73	−7,954	−8,196	−7,760
GPRO	MA	299	−56.62	−56.62	−57.24	−682,477	−682,477	−690,382
	OBV	150	−26.52	−26.52	−26.10	−310,935	−310,935	−305,665
	IMB	618	−109.10	−108.87	−109.82	−1,293,808	−1,291,173	−1,301,713
EUR/USD	MA	56	−0.19	−0.19	−0.18	−1,899	−1,945	−1,835
	OBV	158	−1.48	−1.45	−1.36	−14,846	−14,617	−13,695
	IMB	53	−0.27	−0.28	−0.28	−2,730	−2,766	−2,803

- **Every strategy loses money net of the spread.** Total returns range from −0.19% for MA on EUR/USD (56 round trips) to −109% for IMB on GPRO (618 round trips).
- **A book below zero is possible** because the same number of units is traded on every signal without compounding: 618 round trips at about 18 bps each cost more than the book.
- **This is expected.** The paper studies speed, not profitability, and most of its rules lose as well (58% of those that trade on a given day).

5.1 Where the P&L comes from

Table 5. Decomposition of the P&L with instantaneous execution. *P&L at mid* values the same trades at the mid quote, so *Spread paid* is the difference with the actual P&L. *Holding* is the average length of a position; *In pos.* is the share of the session spent long or short.

Asset	Signal	P&L at mid (\$)	Spread paid (\$)	Bps/trip	Wins (%)	Holding (min)	In pos. (%)
AAPL	MA	−37,878	23,086	−1.9	33	13.8	57
	OBV	−1,334	8,269	−0.9	42	12.2	17
	IMB	−4,971	2,983	−1.8	31	1.3	1
GPRO	MA	+108,037	790,514	−18.9	18	19.1	77
	OBV	+84,321	395,257	−17.7	17	10.7	22
	IMB	+339,921	1,633,729	−17.7	9	3.8	32
EUR/USD	MA	−347	1,552	−0.3	27	9.8	7
	OBV	−10,564	4,282	−0.9	23	9.6	20
	IMB	−1,516	1,214	−0.5	25	1.8	1

- **On GPRO the loss is the spread.** Valued at the mid, the three GPRO strategies make money: +\$108,037 for MA, +\$84,321 for OBV and +\$339,921 for IMB. The spread paid is \$0.4 to \$1.6 million. The signals predict the direction of the next move, but not by the 22.5 bps a round trip costs.
- **On AAPL the signal itself loses.** AAPL MA loses \$37,878 at the mid out of \$60,963 in total: a trend rule on one-minute mids buys after the rise and the price partly reverts.
- **Holding times.** MA positions last 10 to 19 minutes on average, OBV positions 10 to 12 minutes, IMB positions 1 to 4 minutes.
- **Trading frequency.** MA trades about 300 times on each stock and 56 times on EUR/USD, whose one-minute volatility (1.1 bps) rarely carries the short average 5 bps away from the long one. IMB trades far more on GPRO (618 round trips) than on AAPL (42) or EUR/USD (53): GPRO's queues are large and persistently one-sided.

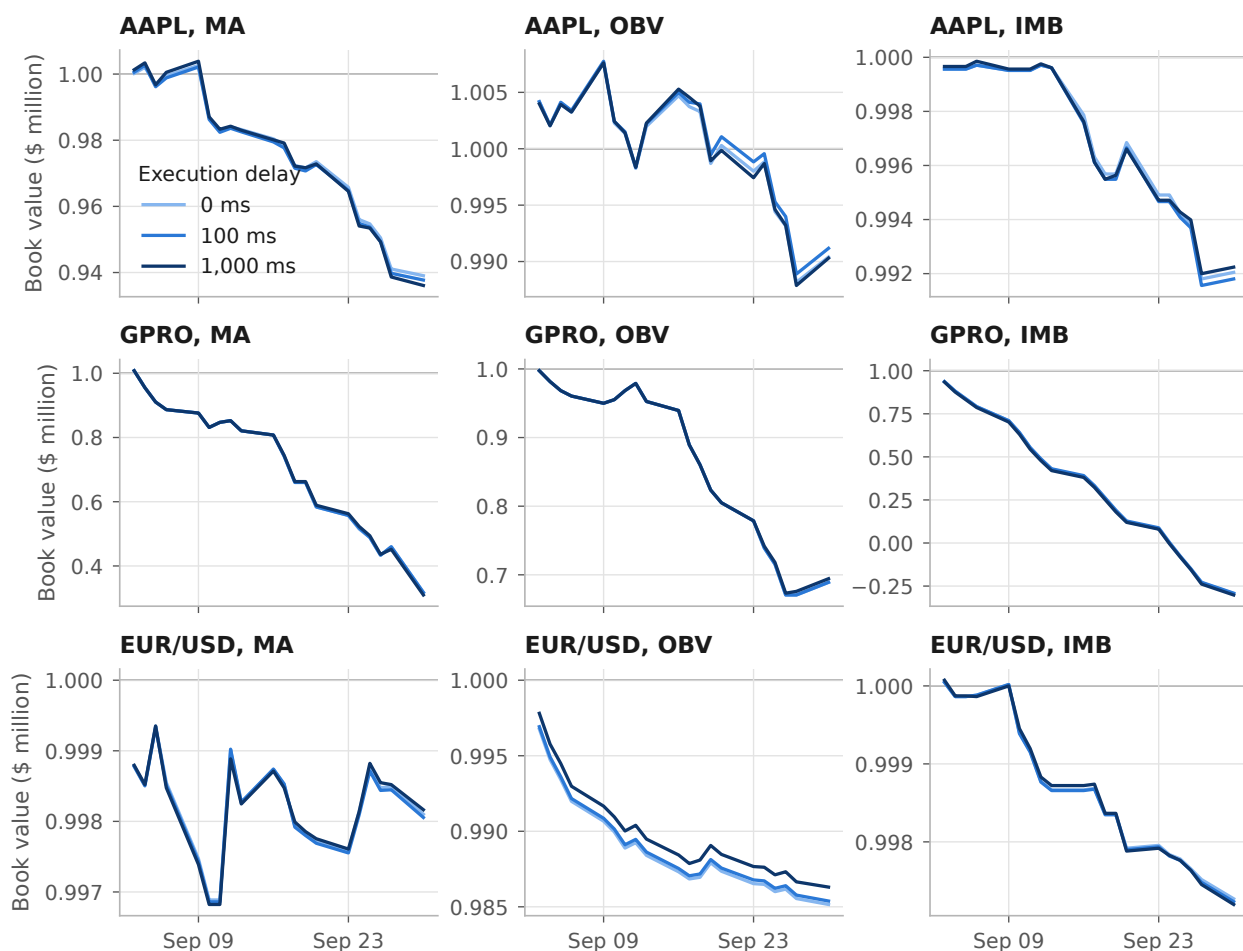


Figure 2. Book value of the 27 strategies over the month, starting from \$1,000,000. In each panel the three delays are almost on top of each other.

6 Cost of delay of the nine strategies

Key finding. For a single strategy over one month, the cost of delay is small and not statistically distinguishable from zero: +0.02% at 100 ms and −0.12% at 1 second on average over the nine asset/signal pairs.

Table 6 gives the cost of delay at the two required delays, with two measures of uncertainty: the Wilcoxon signed-rank test on the 20 daily differences, which is the paper’s test, and a 90% interval obtained by resampling the 20 trading days 5,000 times. Figure 3 extends the delays to the whole range from 0 to 1,000 ms.

Table 6. Cost of delay of the nine strategies: total return with the delay minus total return at 0 ms, in percentage points. The Wilcoxon test is not reported when fewer than six days differ.

Asset	Signal	100 ms		1 second			Wilcoxon <i>p</i>
		Cost (%)	90% interval	Cost (%)	90% interval	Cost (\$)	
AAPL	MA	−0.12	[−0.39, 0.09]	−0.27	[−0.80, 0.26]	−2,958	0.37
	OBV	+0.07	[0.01, 0.14]	−0.01	[−0.20, 0.18]	−97	0.90
	IMB	−0.02	[−0.08, 0.02]	+0.02	[−0.05, 0.09]	+194	0.54
GPRO	MA	0.00	[−0.44, 0.44]	−0.62	[−2.31, 0.68]	−7,905	0.81
	OBV	0.00	[0.00, 0.00]	+0.42	[0.00, 0.87]	+5,270	n/a
	IMB	+0.23	[−0.40, 0.86]	−0.72	[−2.16, 0.57]	−7,905	0.28
EUR/USD	MA	−0.005	[−0.015, 0.007]	+0.006	[−0.026, 0.043]	+64	0.85
	OBV	+0.023	[0.002, 0.053]	+0.115	[−0.015, 0.322]	+1,150	0.57
	IMB	−0.004	[−0.007, −0.001]	−0.007	[−0.026, 0.013]	−73	0.45
Mean of the nine		+0.02	[−0.07, 0.10]	−0.12	[−0.42, 0.14]		

Reading the table.

- **At 100 ms the cost is within 0.23 percentage points for all nine pairs**, with no consistent sign: negative for AAPL MA (−0.12%), positive for GPRO IMB (+0.23%) and AAPL OBV (+0.07%), zero for GPRO MA and OBV, whose execution prices almost never change within 100 ms.
- **At 1 second the less liquid stock pays the most.** GPRO IMB loses 0.72 percentage points and GPRO MA 0.62 (−\$7,905 each). AAPL MA loses 0.27 (−\$2,958) and EUR/USD MA is unchanged.
- **The OBV strategies gain from delay** at 1 second on GPRO (+0.42%) and EUR/USD (+0.12%).
- **No 1-second cost is statistically distinguishable from zero.** Every 1-second interval contains zero or touches it. On GPRO a single one-cent difference on one execution moves the total return by 0.22%.

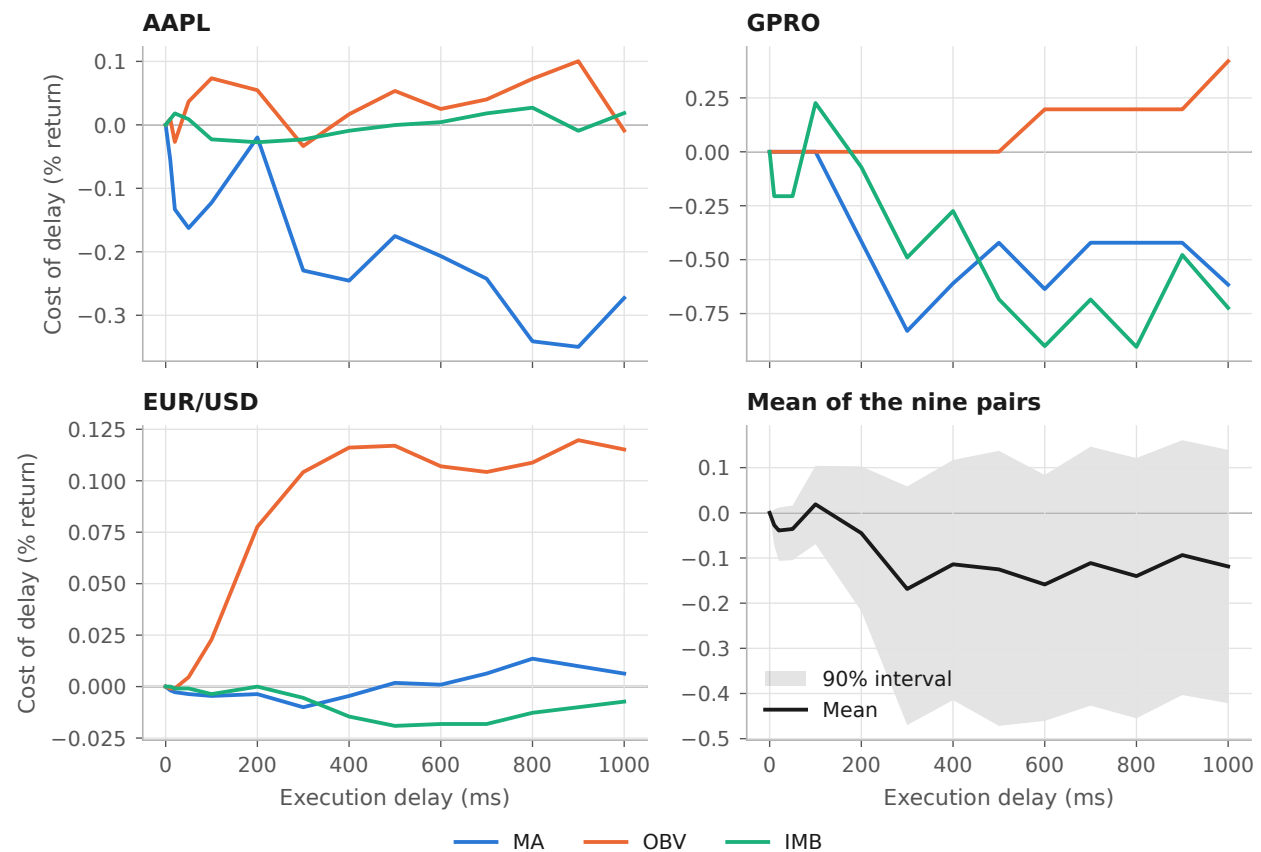


Figure 3. Cost of delay from 0 to 1,000 ms for each asset and signal, and mean of the nine pairs with its 90% interval.

Reading the figure.

- **The mean curve is negative at every delay from 200 ms**, between -0.05% and -0.17% . This is the shape of the discussion-session example: a small, noisy effect at short delays, a steady cost at longer ones.
- **Its interval contains zero at every delay:** -0.42% to $+0.14\%$ at 1 second.

Why the next section is needed. One strategy over one month places 84 to 1,236 orders. That is not enough to measure an effect of a few hundredths of a basis point per order, which is why the paper works with thousands of rules.

7 Importance of speed over a universe of 601 rules

This section repeats the measurement over 601 rules per asset: 288 MA rules, 288 OBV rules and 25 IMB rules (Appendix A). It follows the paper in three steps: all rules together, then rules that make money and rules that lose money with a test of the selection bias, and finally a measure per order that we add.

7.1 All rules (Figure 6 of the paper)

As in the paper, each day the average return across rules with delay d is compared with the average under instantaneous execution, in percent of the latter, and the daily figures are averaged over the 20 days. Rules with a zero return on a day are excluded; on an average day 414 rules trade on AAPL, 547 on GPRO and 303 on EUR/USD.

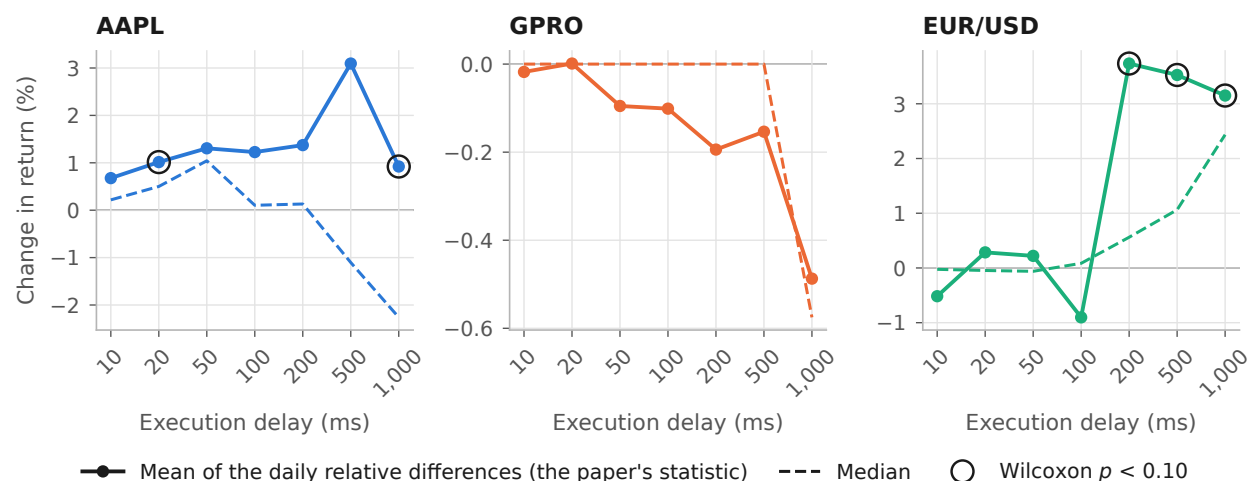


Figure 4. Importance of speed for all rules, at the paper's seven delay levels.

Table 7. Importance of speed for all rules. The first statistic is the paper's (mean over days of the relative change in the average return); the second is the median of the same daily figures; the third is the change in the average return in bps per rule-day, which does not depend on a ratio, with its Wilcoxon test.

Asset	Delay (ms)	Mean of daily ratios (%)	Median (%)	Bps per rule-day	Wilcoxon p
AAPL	100	+1.23	+0.10	0.00	0.60
	200	+1.37	+0.13	-0.08	0.93
	500	+3.09	-1.10	-0.32	0.35
	1,000	+0.92	-2.26	-0.77	0.064
GPRO	100	-0.10	0.00	-0.16	0.64
	200	-0.19	0.00	-0.29	0.28
	500	-0.15	0.00	-0.21	0.84
	1,000	-0.49	-0.58	-0.86	0.11
EUR/USD	100	-0.90	+0.08	+0.01	0.81
	200	+3.74	+0.56	+0.08	0.004
	500	+3.53	+1.06	+0.12	0.004
	1,000	+3.15	+2.43	+0.17	0.002

- At 100 ms nothing is distinguishable from zero on any asset.

- **GPRO declines with delay:** -0.10% at 100 ms and -0.49% at 1 second, or 0.86 bps per rule-day ($p = 0.11$).
- **AAPL loses at 1 second, but the paper's statistic hides it.** The mean of daily ratios is positive ($+0.9\%$) because a few days on which the average return is close to zero dominate it. The median is -2.3% , and in return terms a 1 second delay costs 0.77 bps per rule-day ($p = 0.06$).
- **On EUR/USD delay helps from 200 ms:** $+0.17$ bps per rule-day at 1 second ($p = 0.002$).

These averages mix rules that make money with rules that lose money. The paper shows that the two behave differently, and warns that separating them creates a bias.

7.2 Winning and losing rules, and the selection bias (Figure 7 of the paper)

Key finding. The paper's central result holds on AAPL: delay lowers the return of profitable rules by 0.7% at 200 ms and 3.1% at 1 second, more than selection alone can explain. On GPRO and EUR/USD the losses of similar size are what selection alone produces.

The bias. Selecting the rules with a positive return at 0 ms selects trades whose first instants went well. Delaying them removes part of that return mechanically, even if the signal contains no information.

The test. The paper measures this bias by applying the signals generated on one day to the prices of another, random day. These random rules trade as often and hold as long as the real ones, but know nothing about the day they trade. We do the same 5,000 times. An actual value below the 5th percentile of the random rules is a loss that selection cannot explain: a one-sided test at the 5% level, as in the paper.

Table 8 gives the result at four delays, and Figure 5 shows the seven delay levels.

Table 8. Importance of speed (%) for winning and losing rules. *Random rules* is the 5th to 95th percentile of the same statistic for the random rules; p is the share of the 5,000 draws at or below the actual value. A star marks $p < 0.05$.

Asset	Delay (ms)	Winning rules			Losing rules		
		Actual (%)	Random rules	p	Actual (%)	Random rules	p
AAPL	100	-0.46	[-0.49, 0.61]	0.058	+0.23	[0.17, 0.93]	0.091
	200	-0.73*	[-0.61, 0.80]	0.034	+0.23	[0.23, 1.25]	0.052
	500	-0.61*	[-0.48, 1.48]	0.030	-0.75*	[-0.09, 1.45]	0.001
	1,000	-3.13*	[-1.41, 1.68]	< 0.001	-1.67*	[-0.50, 1.73]	< 0.001
GPRO	100	+0.01	[-0.32, 1.27]	0.42	-0.09*	[-0.07, 0.21]	0.034
	200	-0.01	[-0.51, 1.30]	0.39	-0.16*	[-0.13, 0.21]	0.033
	500	-0.31	[-5.81, 2.30]	0.54	-0.11	[-0.27, 0.29]	0.27
	1,000	-0.96	[-5.55, 2.57]	0.47	-0.32	[-0.38, 0.28]	0.087
EUR/USD	100	+0.06	[-0.96, 0.07]	0.94	-0.08	[-0.15, 0.37]	0.11
	200	+0.04	[-1.75, 0.25]	0.89	+0.65	[0.22, 1.34]	0.41
	500	-0.66	[-3.00, 0.32]	0.69	+1.24	[0.26, 2.02]	0.62
	1,000	-1.23	[-4.31, -0.31]	0.76	+1.66	[0.08, 2.33]	0.76

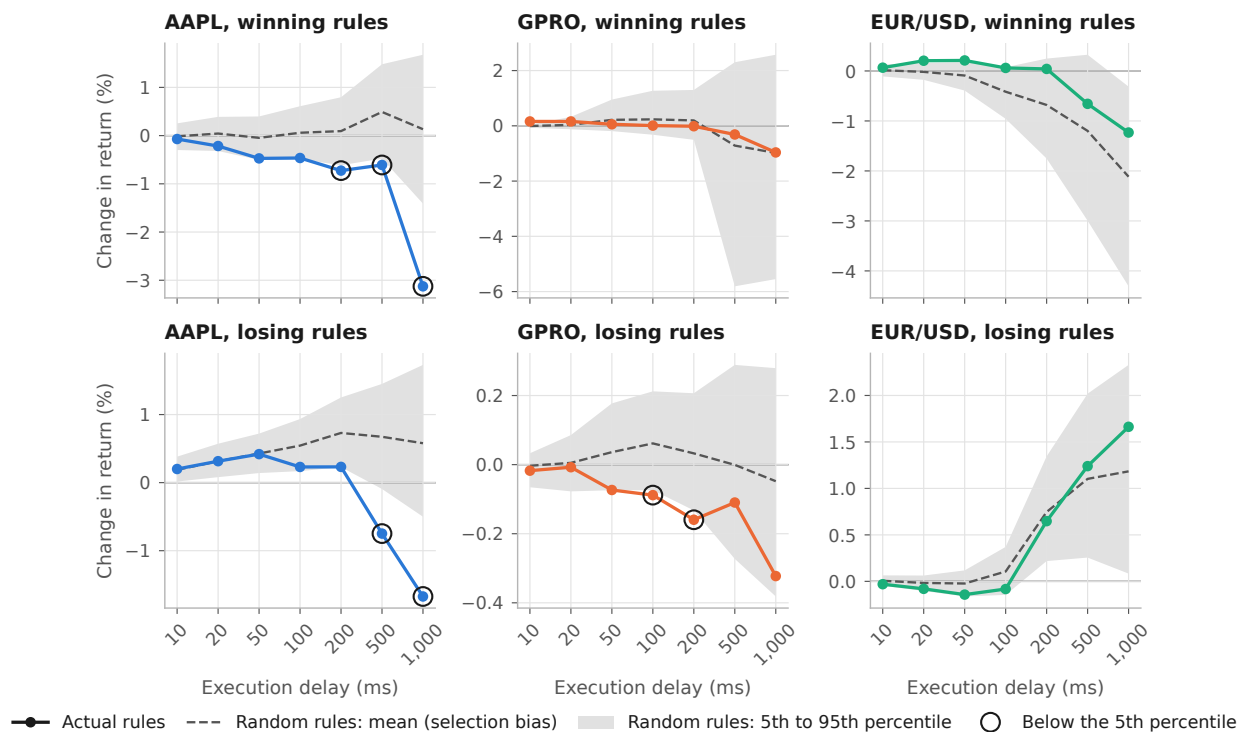


Figure 5. Importance of speed for rules with a positive return at 0 ms (top) and a negative return (bottom), against the same statistic for the random rules.

- **AAPL, winning rules: speed matters.** Delay lowers their return by 0.5% at 100 ms, 0.7% at 200 ms and 3.1% at 1 second. Random rules lose nothing (+0.1% at 1 second), so this is not a selection effect: $p = 0.06$ at 100 ms, 0.03 at 200 ms and at 500 ms, below 0.001 at 1 second. This is the paper's main result, with the same threshold of about 200 ms.
- **AAPL, losing rules.** Up to 200 ms they behave like random rules. At 500 ms and 1 second they lose more (−1.7% at 1 second against +0.6% for random rules). The paper finds no effect of speed on losing rules; on AAPL a 1 second delay hurts whatever the rule's result.
- **GPRO, winning rules: no evidence.** The −1.0% at 1 second equals the selection bias. Only 67 rules a day make money on GPRO, 12% of those that trade, so the band is wide. Losing rules lose slightly more than random ones at 100 and 200 ms.
- **EUR/USD: the raw numbers are selection.** Winning rules lose 1.2% at 1 second, but random rules lose 2.1%. Losing rules gain 1.7%, and random ones gain 1.2%. Neither is evidence of an effect of speed.

Without the random rules we would have concluded that delay hurts winning rules on all three assets. Beyond selection, it only does on the liquid stock.

7.3 Comparison with the paper

Table 9. Importance of speed (%) for rules with a positive return. Paper: Table B.1 of its internet appendix (186 days, 27,424 rules). A star marks a loss that is significant against the selection bias.

Delay (ms)	Paper, Jan–Sep 2009			This project, Sep 2019		
	SPY	QQQQ	IWM	AAPL	GPRO	EUR/USD
100	−0.07	−0.07	−0.07	−0.46	+0.01	+0.06
200	−0.22*	−0.26	−0.15	−0.73*	−0.01	+0.04
500	−0.39*	−1.01*	−0.62*	−0.61*	−0.31	−0.66
1,000	−1.05*	−1.91*	−1.34*	−3.13*	−0.96	−1.23

The liquid stock reproduces the paper: no significant effect at 100 ms, a significant loss from 200 ms, and a loss at 1 second (−3.1%) of the same order as the paper’s ETFs (−1.0% to −1.9%). Ten years later, and on a single stock instead of an index ETF, the importance of speed has the same shape. On the less liquid stock and the currency pair, 20 days cannot separate losses of a similar size from the selection bias.

7.4 Effect of the delay per order

Key finding. Measured order by order, a 1 second delay costs 0.06 to 0.11 bps on the two stocks. A delay of 100 ms costs nothing measurable on any asset.

The split between winning and losing rules conditions on performance. The following measure does not. For every order of every rule, the effect of executing it h later is

$$e_h = -\delta \frac{p_{t+h} - p_t}{p_t},$$

with $\delta = +1$ and p the ask for a buy, $\delta = -1$ and p the bid for a sell. It is negative when the price moves in the direction of the order during the delay. Summed over the orders of a strategy, it is that strategy’s cost of delay.

Table 10. Effect of the delay per order, in bps of the price, over the 601 rules. Intervals resample trading days, which keeps the dependence between rules that trade at the same instant. A star marks a 90% interval that excludes zero.

Asset	Signal	Orders	100 ms		1 second	
			Bps / order	90% interval	Bps / order	90% interval
AAPL	MA	83,616	−0.002	[−0.012, 0.007]	−0.070*	[−0.113, −0.032]
	OBV	59,390	−0.001	[−0.013, 0.010]	−0.027	[−0.074, 0.012]
	IMB	11,702	+0.006	[0.000, 0.011]	+0.015	[−0.009, 0.037]
GPRO	MA	121,602	−0.006	[−0.025, 0.010]	−0.058*	[−0.109, −0.007]
	OBV	78,714	−0.007	[−0.031, 0.017]	+0.003	[−0.045, 0.056]
	IMB	25,622	−0.017	[−0.043, 0.005]	−0.107*	[−0.180, −0.040]
EUR/USD	MA	39,138	−0.0005	[−0.0019, 0.0009]	+0.0029	[−0.0026, 0.0086]
	OBV	90,128	+0.0010	[−0.0002, 0.0023]	+0.0104*	[0.0061, 0.0151]
	IMB	9,438	−0.0011*	[−0.0020, −0.0003]	−0.0021	[−0.0058, 0.0016]

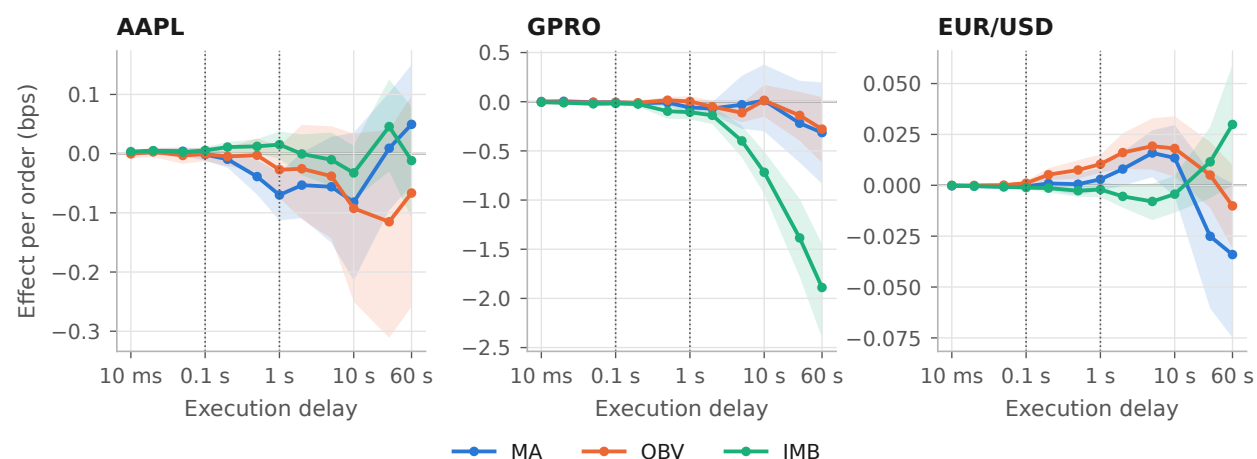


Figure 6. Effect of executing an order later, from 10 ms to 60 seconds, by signal family. Bands are 90% intervals; the dotted lines mark 100 ms and 1 second.

- **100 ms costs nothing measurable.** Every family on every asset is within 0.02 bps per order, and only one interval excludes zero (EUR/USD IMB, -0.001 bps).
- **1 second has a cost on the two stocks:** 0.07 bps per order for MA rules on AAPL, 0.06 bps for MA rules on GPRO and 0.11 bps for IMB rules on GPRO.
- **The cost is the signal's short-run predictive power.** After an IMB order on GPRO the price keeps moving in the order's direction: 0.4 bps after 5 seconds, 1.9 bps after 60 seconds. Each second of delay gives up a slice of that move. On AAPL the drift after MA orders builds up over the first second and is gone after 30 seconds.
- **Relative to the spread the cost is small.** 0.07 bps is 19% of AAPL's half-spread; 0.11 bps is 1% of GPRO's. On EUR/USD delay even helps OBV rules, by 0.01 bps per order at 1 second.

8 Answers to the questions of the assignment

8.1 Question (a): which asset is affected by delays, and why?

Key finding. The two stocks are affected at 1 second and the currency pair is not. Nothing is affected at 100 ms.

- **AAPL is the asset where speed matters most clearly.** Its book changes 8.8 times per second, so a delayed order usually meets a different quote, and a one-cent move is worth more than its half-spread. The cost is significant from 200 ms for profitable rules.
- **GPRO pays the most in percent for a single strategy.** Its MA and IMB strategies lose 0.6% and 0.7% at 1 second, against 0.3% for AAPL's MA, because each change of its quote is a full tick of 22 bps. But its quote changes so rarely that the cost comes from a handful of executions, and it is not statistically measurable for one strategy.
- **EUR/USD is not affected.** Its ticks are frequent but tiny (under 0.1 bp), and its one-minute signals have no short-run predictive power to lose.

Why. A delay costs money when two conditions hold: the quote moves during the delay, and it moves in the direction of the order, which requires a signal that predicts the next move. On AAPL both hold within a second for the MA rules. On GPRO the quote rarely moves, but the imbalance signal predicts its direction when it does. On EUR/USD the quote moves often, by very little, and not in the direction of the orders.

8.2 Question (b): do the signals depend on market conditions?

Key finding. Yes. The cost of delay is concentrated on volatile days, and it is carried by the MA rules on both stocks and by the IMB rules on GPRO.

Days are split, asset by asset, into the 10 most and 10 least volatile, by one-minute mid-quote volatility.

Table 11. Signals across volatility regimes, over the rule universe.

Asset	Signal	Return at 0 ms (bps / rule-day)		Cost of 1 s (bps / rule-day)		Effect of 1 s per order (bps)	
		High vol	Low vol	High vol	Low vol	High vol	Low vol
AAPL	MA	−27.4	−18.6	−2.14	−0.34	−0.111	−0.017
	OBV	−9.7	−11.2	−1.27	+0.44	−0.080	+0.027
	IMB	−9.8	−18.2	−0.13	+1.43	−0.003	+0.032
GPRO	MA	−256.1	−246.1	−3.16	+0.65	−0.135	+0.032
	OBV	−157.3	−161.4	+0.07	+0.02	+0.004	+0.001
	IMB	−517.7	−406.0	−9.70	−1.40	−0.174	−0.029
EUR/USD	MA	−5.1	−7.3	+0.05	+0.08	+0.0027	+0.0032
	OBV	−9.1	−7.6	+0.33	+0.15	+0.0137	+0.0069
	IMB	−5.8	−6.4	−0.13	−0.01	−0.0039	−0.0002

- **MA.** A 1 second delay costs MA rules 0.11 bps per order on AAPL's volatile days against 0.02 on quiet days, and 0.14 bps on GPRO's volatile days against a small gain on quiet days. The quote moves further during the delay when the market moves faster.
- **IMB.** On GPRO it loses 0.17 bps per order to a 1 second delay on volatile days against 0.03 on quiet days.
- **OBV changes sign.** On AAPL a 1 second delay costs OBV rules 0.08 bps per order on volatile days and helps them by 0.03 on quiet days. On GPRO it shows nothing in either regime.
- **Performance itself depends on the regime.** MA rules on AAPL lose more on volatile days (−27 against −19 bps per rule-day): the band is crossed more often and reversed more often. IMB on GPRO also loses more (−518 against −406), because it trades more and pays the spread each time. IMB on AAPL loses less (−10 against −18).
- **This differs from the paper,** which finds that speed matters more on low-volatility days for SPY.

8.3 Question (c): does tick frequency matter, and how to use it?

Key finding. Partly. Tick frequency sets the probability that a delayed order meets a stale quote; tick size relative to price sets what a stale quote costs. Sensitivity to delay is the product of the two.

Table 12. Tick frequency and sensitivity to delay. The absolute effect is the mean absolute change in return caused by the delay, per round trip, over the rule universe; it measures sensitivity regardless of sign.

	AAPL	GPRO	EUR/USD
Quote price changes per second	8.80	0.04	1.21
Quote changed after 100 ms / 1 s (%)	25.6 / 62.2	0.2 / 1.1	15.0 / 55.0
Mean absolute mid move after 1 s (bps)	0.48	0.24	0.08
in half-spreads	1.31	0.02	0.62
Absolute effect of 100 ms per round trip (bps)	0.12	0.08	0.01
Absolute effect of 1 s per round trip (bps)	0.38	0.42	0.07
Correlation of daily sensitivity with quote changes	0.48	0.57	0.55
Correlation of daily sensitivity with volatility	0.48	0.69	0.67

- **At 100 ms the fast book is the most exposed:** the absolute effect per round trip is 0.12 bps on AAPL, against 0.08 on GPRO and under 0.02 on EUR/USD.
- **At 1 second GPRO is as sensitive as AAPL** (0.42 against 0.38 bps), although its book is 200 times slower, because each of its ticks is worth 22 bps. EUR/USD is far behind (0.07 bps), although it ticks 30 times faster than GPRO.
- **Within an asset, busier days are more delay-sensitive:** the correlation of daily sensitivity with the number of quote changes per second is 0.48 to 0.57.

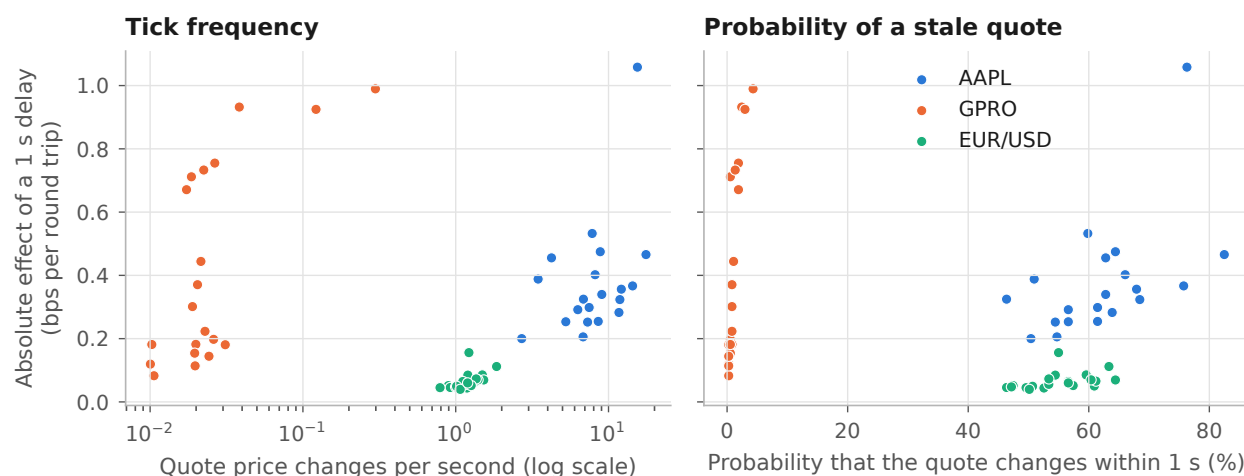


Figure 7. Sensitivity to a 1 second delay against tick frequency. One point per asset-day.

How we would factor this into a trading strategy.

1. **Budget latency per asset in units of expected quote move, not in milliseconds.** The relevant number is the probability that the quote changes during the delay times the size of a tick in bps, compared with the signal's expected edge. On AAPL a quarter of the quotes have changed after 100 ms and the measured cost starts around 200 ms; on GPRO 100 ms is free.
2. **Match the signal's horizon to the achievable latency.** IMB on GPRO pays out over a minute, so a 1 second delay gives up about 6% of its move. The drift after MA orders on AAPL is over within a second: that edge exists only for a sub-second trader.
3. **In large-tick, slow books, do not cross the spread.** GPRO's signals predict about 2 bps and a round trip costs 22.5 bps. The same signals should be expressed with passive orders: queue position then replaces latency as the scarce resource.
4. **Do not pay for speed on a signal that loses.** Beyond selection, the cost of delay falls on AAPL's profitable rules. Speed is valuable once the signal predicts the next move; otherwise it just delivers the wrong trade sooner.
5. **Scale speed requirements with volatility.** The cost of a 1 second delay is concentrated on volatile days for both stocks; wider bands or standing aside around scheduled events limit the exposure when the infrastructure cannot be faster.

9 Conclusion and limitations

9.1 Conclusion

Over one month and three very different markets, the cost of execution delay is real but small. At 100 ms we measure nothing. At 1 second we measure a cost of 0.06 to 0.11 bps per order on the two stocks, and on AAPL the paper's main result holds after its selection-bias test: delay lowers the return of profitable rules from about 200 ms. The cost is always far below the bid–ask spread, which is what makes all 27 strategies lose money. Speed is therefore worth paying for only when the signal predicts the next move and the strategy can afford the spread, and the latency that matters depends on the asset: tick frequency decides how often a delayed order meets a different quote, and tick size decides what that costs.

9.2 Limitations

- **One month.** Single-strategy costs of delay cannot be measured, and the pooled results rest on 20 days. With seven delays, three assets and two groups of rules, a few rejections at the 5% level are expected by chance; the AAPL results at 1 second ($p < 0.001$) do not depend on that.
- **Fill at the best quote.** The order is 22 and 53 times the displayed size on the two stocks. True costs, and the cost of arriving after others have taken the best level, are understated there.
- **EUR/USD.** One venue's quotes, tick volume in place of traded volume, New York hours only.
- **Nasdaq quotes only.** The consolidated best bid and offer may be tighter than the Nasdaq book for AAPL (1.6 cents on average here). GPRO is already at the one-cent minimum.
- **Start of the day.** Lookbacks start at the open, so rules with a 30-minute lookback trade from 10:00 and those with a 60-minute lookback from 10:30.
- **No fees or rebates,** as in the paper.

Appendix A Rule universe

Table 13. Rules of the universe. The lookbacks and the MA bands are the paper’s 60-second settings; its two universal filters are at their neutral values (no confirmation delay, no minimum holding period).

Family	Parameters	Rules
MA	$s \in \{2, 5, 10, 15, 20, 25, 30\}$, $l \in \{10, 15, 20, 25, 30, 35, 40, 45, 60\}$ $s < l$ (48 pairs), $b \in \{0, 5, 10, 15, 25, 40\}$ bps	288
OBV	the same 48 pairs (s, l) $b \in \{0, 0.1, 0.25, 0.5, 1, 1.5\}$ times the average one-minute volume	288
IMB	$k \in \{1, 2, 3, 4, 5\}$, $\theta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$	25
Total per asset		601

The three baseline rules behind the 27 reports are MA with $(s, l, b) = (5, 30, 5 \text{ bps})$, OBV with $(s, l, b) = (5, 30, 0.5)$ and IMB with $(k, \theta) = (2, 0.3)$.

Appendix B Code and reproduction

Everything is in `Project/Part1_Speed/` of the group repository.

- `MFE230X_project_speed.ipynb`: the executed analysis, with every table and figure of this report.
- `src/speed.py`: data loaders, the three signals, the delayed-execution simulator, the random-rule test and the reporting functions.
- `src/report_assets.py`: writes the tables and figures of this report from the notebook’s outputs.
- `src/download_fx.py`: downloads the EUR/USD ticks.
- `data/`: the Databento and Dukascopy files. `outputs/`: the result tables (CSV) and the notebook’s figures.

To reproduce: install `databento`, `jupyter`, `matplotlib`, `numpy`, `pandas` and `scipy`, execute the notebook from `Part1_Speed/` (about two minutes), then run `python src/report_assets.py`.

Reference. M. Scholtus and D. van Dijk (2012), *High-Frequency Technical Trading: The Importance of Speed*, Tinbergen Institute Discussion Paper.